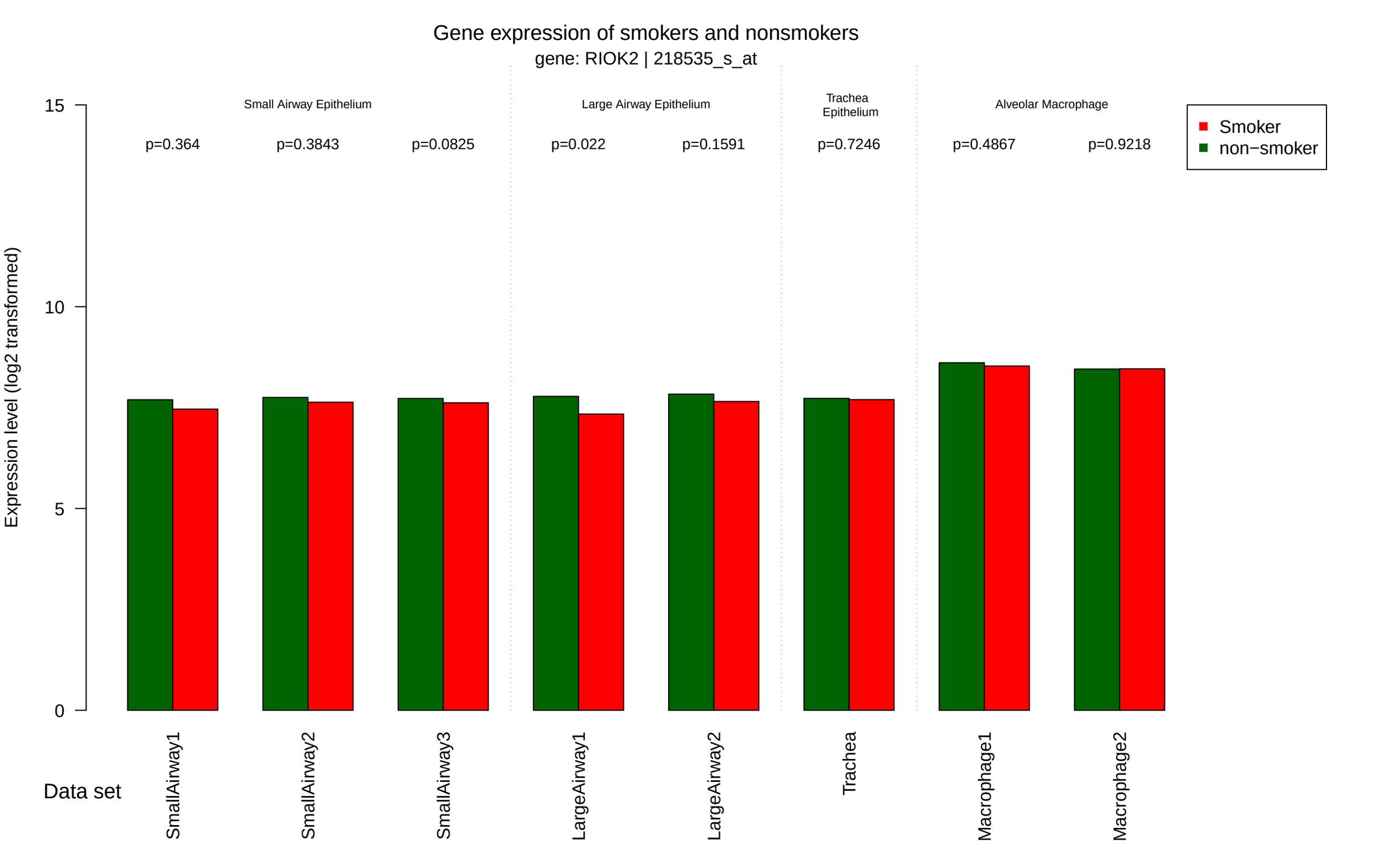
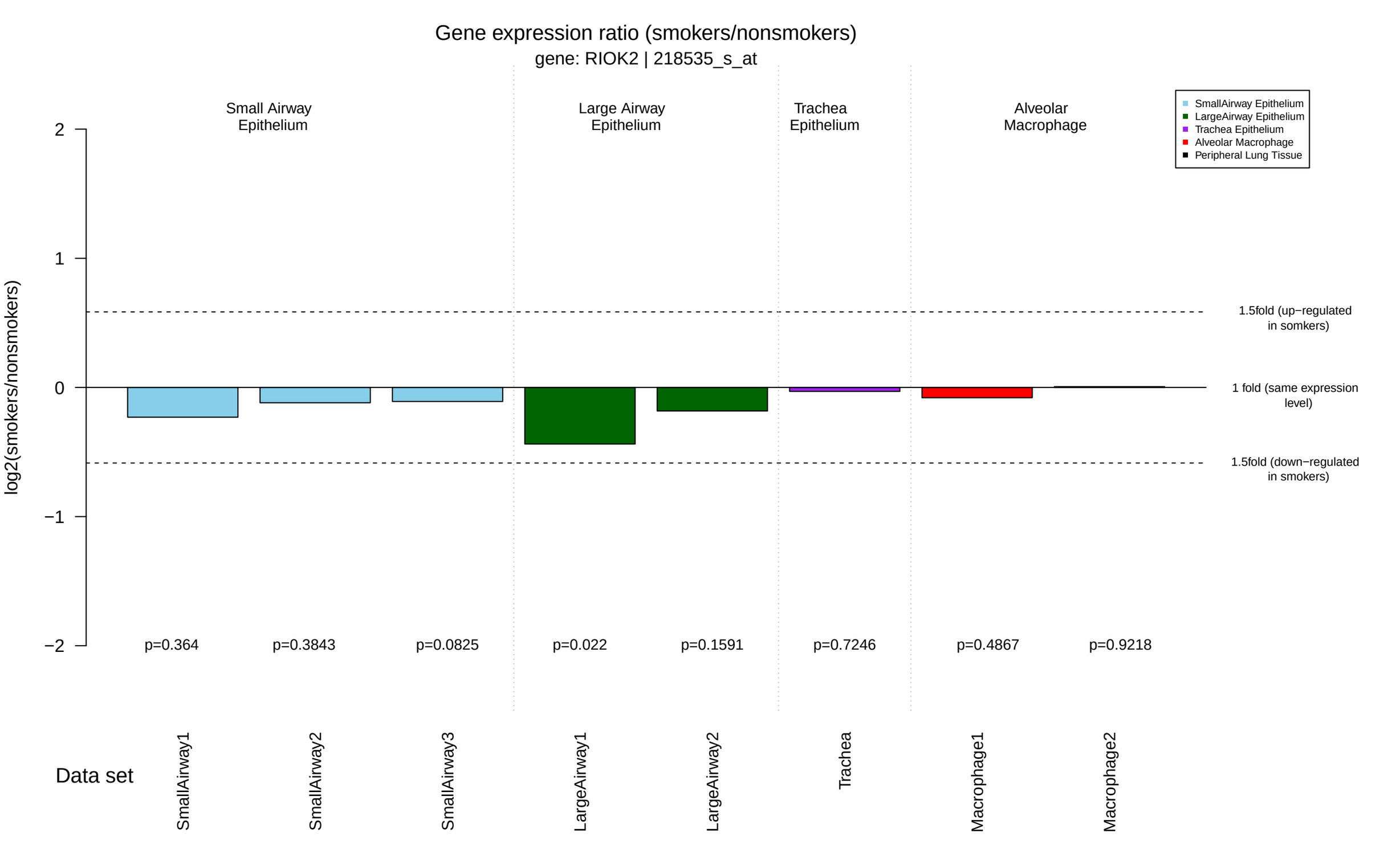
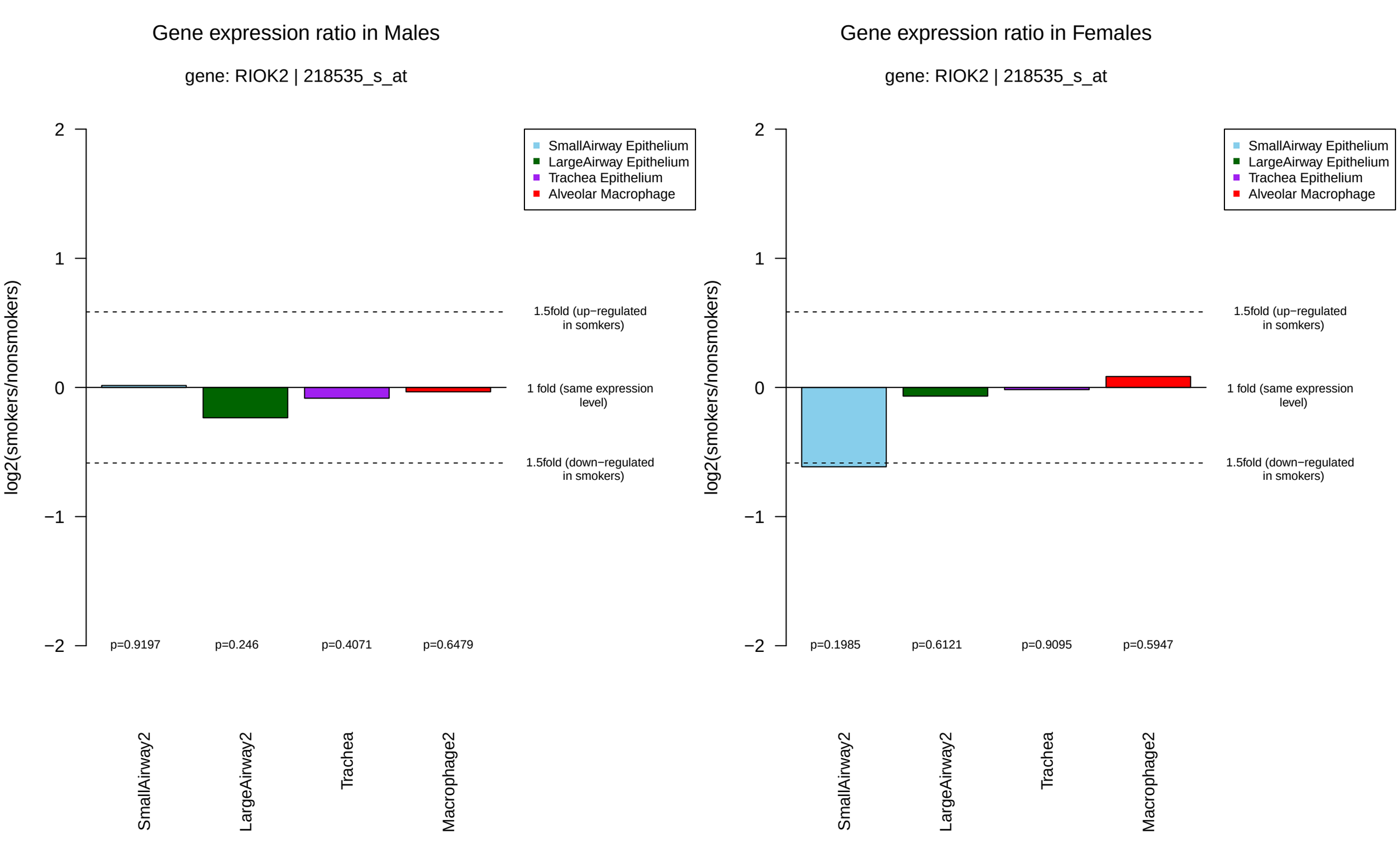
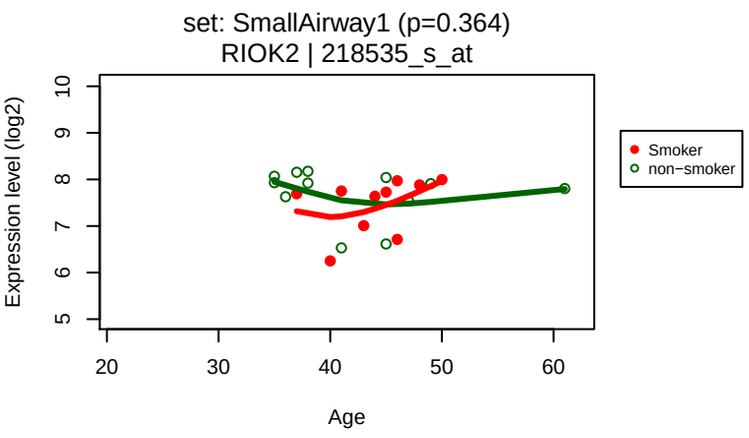
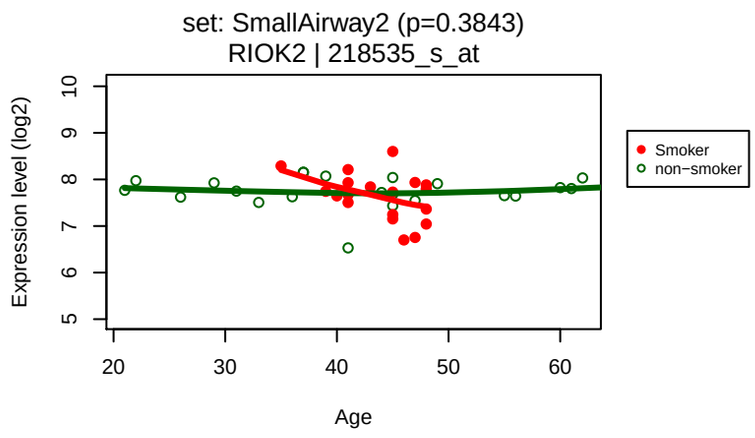
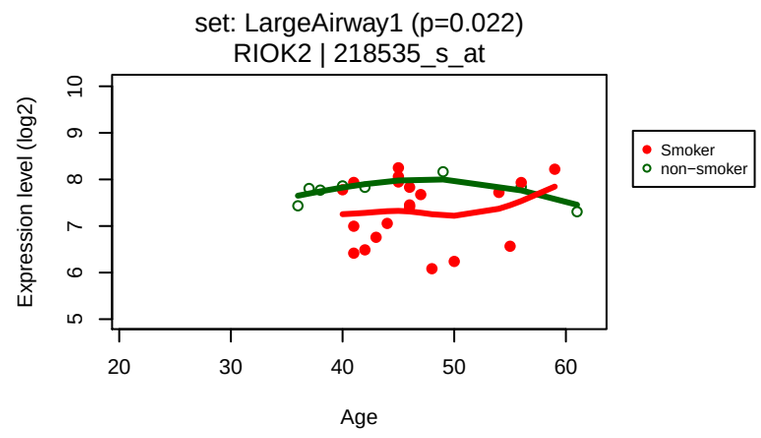

Gene expression of smokers and nonsmokers
gene: RIOK2 | 218535_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.364
p=0.3843
p=0.0825
p=0.022
p=0.1591
p=0.7246
p=0.4867
p=0.9218
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

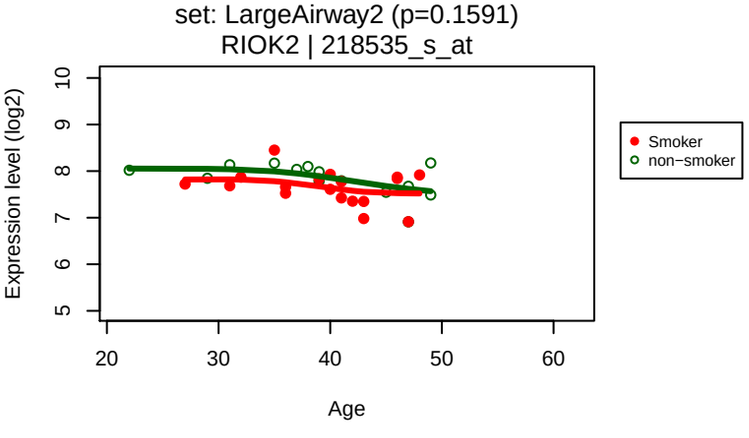

Gene expression ratio (smokers/nonsmokers)
gene: RIOK2 | 218535_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.364
p=0.3843
p=0.0825
p=0.022
p=0.1591
p=0.7246
p=0.4867
p=0.9218
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

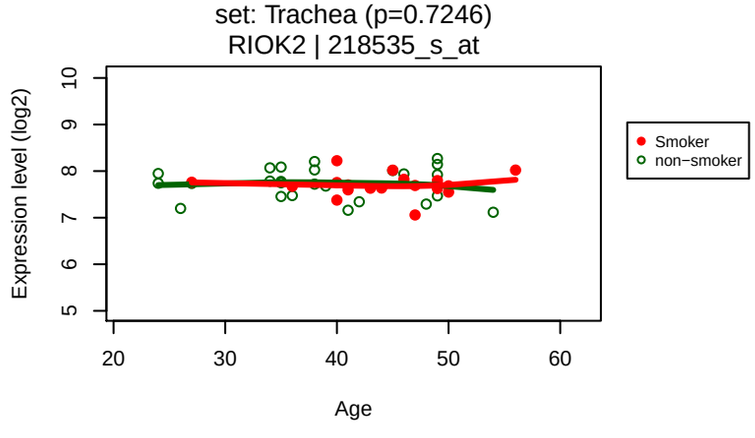

Gene expression ratio in Males
gene: RIOK2 | 218535_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9197
p=0.246
p=0.4071
p=0.6479
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RIOK2 | 218535_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1985
p=0.6121
p=0.9095
p=0.5947
SmallAirway2
LargeAirway2
Trachea
Macrophage2

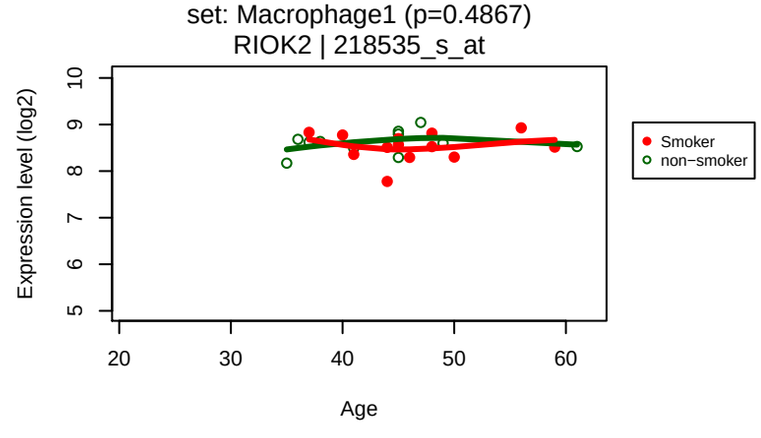

set: SmallAirway1 (p=0.364)
RIOK2 | 218535_s_at
Expression level (log2)
Age
Smoker
non−smoker

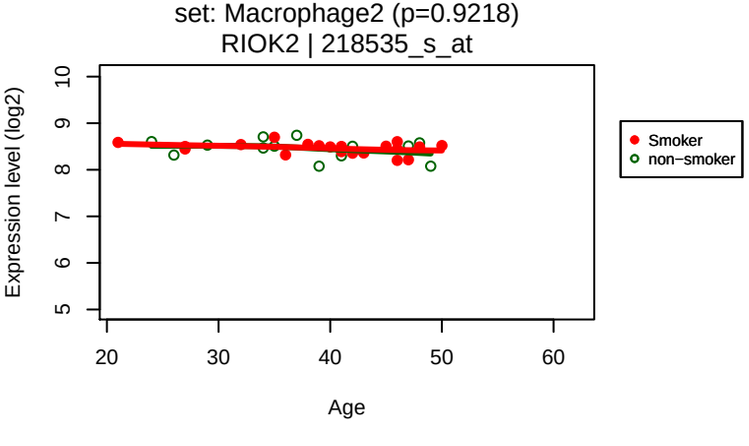

set: SmallAirway2 (p=0.3843)
RIOK2 | 218535_s_at
Expression level (log2)
Age
Smoker
non−smoker

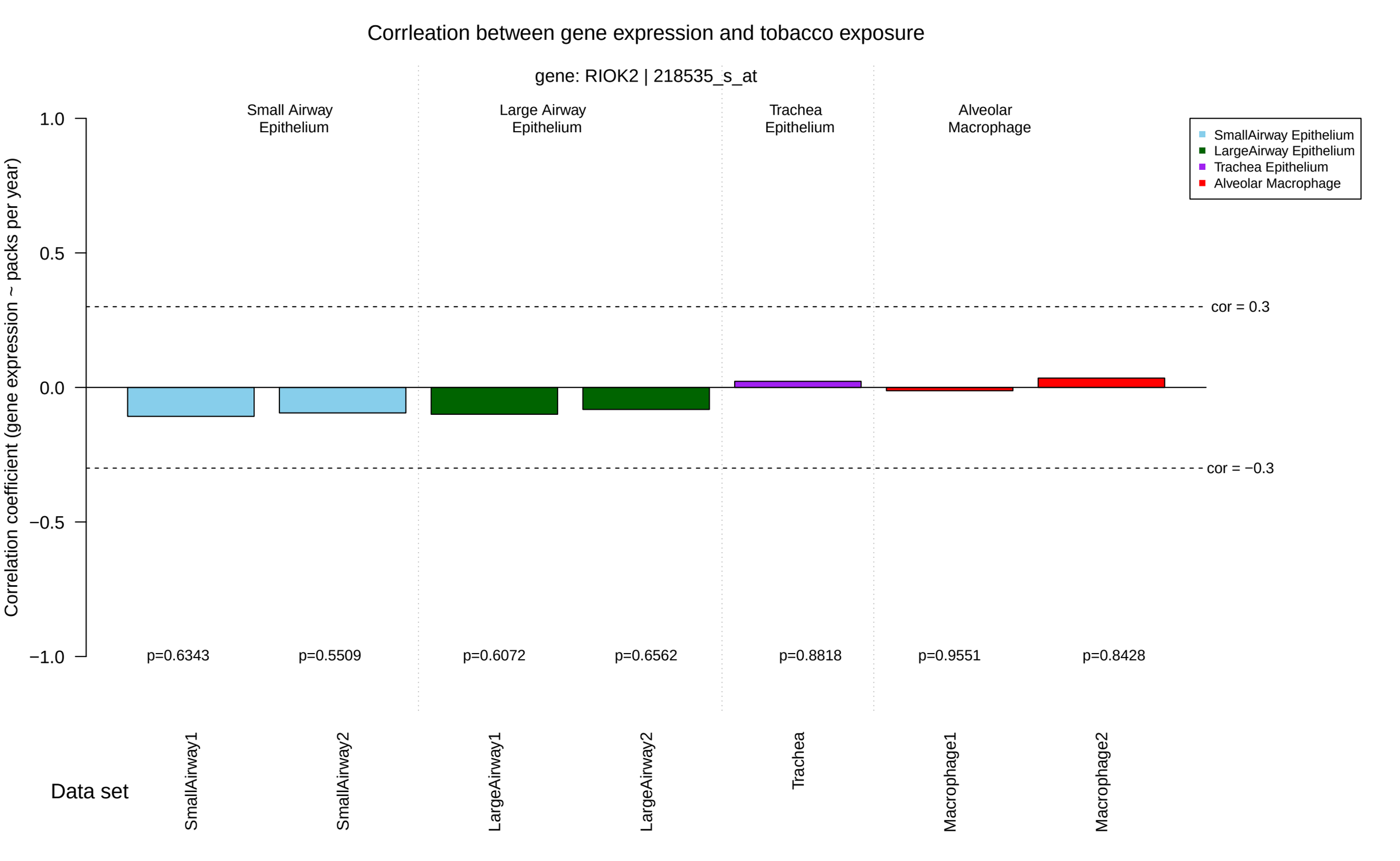

set: LargeAirway1 (p=0.022)
RIOK2 | 218535_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1591)
RIOK2 | 218535_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7246)
RIOK2 | 218535_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4867)
RIOK2 | 218535_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9218)
RIOK2 | 218535_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RIOK2 | 218535_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6343
p=0.5509
p=0.6072
p=0.6562
p=0.8818
p=0.9551
p=0.8428
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2